

BIOL-1 Exam 02

College of the Redwoods, Del Norte

Instructor: Dr. Daniel Friedman

Name: _____

Date: _____

Modules 07-11

Total Points: 50

Part A: Multiple Choice (30 points)

Select the best answer for each question. Each question is worth 1 point.

Module 07: Molecular Genetics

1. Which statement best describes the “central dogma” of molecular genetics in most cells?

- A) Protein → RNA → DNA
- B) RNA → DNA → Protein
- C) DNA → RNA → Protein
- D) DNA → Protein → Lipid

2. In double-stranded DNA, which base pairing is correct?

- A) Adenine (A) pairs with cytosine (C)
- B) Guanine (G) pairs with thymine (T)
- C) Adenine (A) pairs with thymine (T)

- D) Uracil (U) pairs with thymine (T)
3. During transcription, an RNA strand is built using a DNA template. Which pairing rule applies when RNA is being synthesized?
- A) A in DNA pairs with T in RNA
 - B) A in DNA pairs with U in RNA
 - C) G in DNA pairs with T in RNA
 - D) C in DNA pairs with U in RNA only in the cytoplasm
4. Which enzyme builds a new DNA strand during DNA replication by adding nucleotides in the 5'→3' direction?
- A) RNA polymerase
 - B) DNA ligase
 - C) DNA polymerase
 - D) Ribosome
5. Where does translation (protein synthesis from an mRNA code) mainly occur in a typical animal cell?
- A) In the nucleus, on the nuclear envelope
 - B) On ribosomes in the cytoplasm (often on the rough ER)
 - C) Inside the Golgi apparatus
 - D) Inside mitochondria only
6. A single nucleotide is deleted from the coding region of a gene, shifting the downstream reading frame. This is best described as:
- A) A silent point mutation with no possible effect
 - B) A frameshift mutation that often changes many amino acids downstream
 - C) A chromosome duplication
 - D) Epigenetic silencing of the gene

Module 08: Cellular Genetics

1. In humans, a typical body (somatic) cell is diploid ($2n = 46$). A mature sperm or egg cell is:
 - A) Diploid ($2n = 46$)
 - B) Haploid ($n = 23$)
 - C) Triploid ($3n = 69$)
 - D) Haploid ($n = 46$)
2. Mitosis followed by cytokinesis in a diploid human skin cell produces:
 - A) Four genetically different haploid cells
 - B) Two genetically identical diploid daughter cells
 - C) Two genetically identical haploid daughter cells
 - D) One diploid cell and one haploid cell
3. During meiosis I, what separates and moves to opposite poles of the cell?
 - A) Sister chromatids of one chromosome
 - B) Homologous chromosomes (each still made of two sister chromatids)
 - C) Individual replicated DNA bases
 - D) Centrioles only
4. Crossing over (exchange of DNA between homologs) is a major source of genetic variation and occurs during:
 - A) Metaphase of mitosis
 - B) Prophase I of meiosis
 - C) Telophase II of meiosis
 - D) Cytokinesis only in plant cells
5. Independent assortment of homologous chromosomes during metaphase I increases variation because:
 - A) It guarantees identical daughter cells
 - B) Homologs line up randomly, creating many allele combinations in gametes

- C) It prevents crossing over
- D) It copies DNA without errors

6. Nondisjunction refers to:

- A) Normal separation of sister chromatids in mitosis
- B) Failure of chromosomes or chromatids to separate correctly, sometimes producing aneuploidy
- C) The S phase of interphase
- D) Random fertilization of any egg by any sperm

Module 09: Inheritance Genetics

1. In genetics, different versions of the same gene (for example, one version for type A blood and one for type O) are called:

- A) Chromosomes
- B) Alleles
- C) Phenotypes
- D) Mutations

2. For a flower-color gene, B is dominant for purple and b is recessive for white. A plant with genotype Bb has purple flowers. Which statement correctly uses the terms genotype and phenotype?

- A) Bb is the genotype; purple flowers are the phenotype
- B) Bb is the phenotype; purple flowers are the genotype
- C) Purple flowers are both the genotype and the phenotype
- D) Bb is the dominant trait; purple is the recessive trait

3. In a monohybrid cross, two heterozygotes ($Aa \times Aa$) are crossed for a trait with complete dominance. What is the expected phenotypic ratio among offspring?

- A) 1 : 1
- B) 1 : 2 : 1
- C) 3 : 1 (three showing the dominant phenotype, one recessive)

- D) 9 : 3 : 3 : 1
4. In incomplete dominance, a cross of two heterozygotes often produces offspring phenotypes in which ratio?
- A) Only dominant and only recessive (3 : 1)
 - B) 1 : 2 : 1 (two homozygous phenotypes and a blended heterozygote)
 - C) 9 : 3 : 3 : 1
 - D) 4 : 0 (all identical)
5. In the ABO blood system, type AB blood illustrates:
- A) Incomplete dominance (a blended single antigen)
 - B) Codominance (both A and B alleles expressed fully)
 - C) A sex-linked recessive trait
 - D) Polygenic inheritance only
6. Which pattern is a classic clue for an autosomal recessive disorder in a pedigree?
- A) Only males are affected, and an affected father always passes it to sons
 - B) Two unaffected parents can have an affected child
 - C) Every generation must show the trait with no skipping
 - D) The trait never appears in siblings

Module 10: Epigenetics

1. Epigenetics is best defined as:
- A) A change in the DNA sequence (A, T, G, C) of a gene
 - B) Changes in gene expression that do not change the underlying DNA sequence
 - C) Random mistakes during translation only
 - D) The central dogma operating in reverse
2. DNA methylation that silences a gene is often described with the memory trick “Methylation = Muting” because methylation tends to:
- A) Speed up transcription strongly

- B) Turn the gene off or reduce its activity
- C) Delete the gene from the chromosome
- D) Replace thymine with uracil

3. Histone acetylation is associated with looser chromatin and tends to:

- A) Permanently destroy the DNA double helix
- B) Increase access to DNA and favor gene activity (turning genes on)
- C) Prevent all transcription in the cell
- D) Cause nondisjunction

4. Which statement correctly contrasts a mutation with an epigenetic change?

- A) Both always change the DNA sequence
- B) A mutation changes the DNA sequence; epigenetics changes how genes are used without changing the sequence
- C) Epigenetics always changes the amino acid sequence; mutations never do
- D) They are identical terms

5. In female mammals, one X chromosome is inactivated in each somatic cell, forming a dense body called a:

- A) Centromere
- B) Barr body
- C) Okazaki fragment
- D) Plasmid

6. Gene regulation is important because:

- A) Every cell expresses every gene at full strength all the time
- B) Different cell types need different sets of proteins, so cells turn specific genes on or off
- C) DNA is absent from muscle cells
- D) Translation happens only in the nucleus

Module 11: Genomics & Biotechnology

1. PCR (polymerase chain reaction) is used in the lab mainly to:

- A) Destroy all DNA in a sample
- B) Make many copies of a targeted region of DNA
- C) Synthesize only proteins
- D) Replace meiosis in humans

2. In gel electrophoresis, DNA fragments migrate because DNA is negatively charged and move toward:

- A) The negative electrode; larger fragments move farthest
- B) The positive electrode; smaller fragments usually move farther through the gel
- C) The center of the gel only
- D) The top of the gel regardless of charge

3. Moving a human insulin gene into bacteria so the bacteria make human insulin works largely because:

- A) Bacteria do not use DNA
- B) The genetic code is nearly universal, so the bacterium can read the gene and translate it
- C) Bacteria automatically delete foreign genes
- D) Insulin is made of lipids, not proteins

4. Short tandem repeats (STRs) used in DNA fingerprinting vary between individuals and are:

- A) Never inherited from parents
- B) Inherited in Mendelian fashion (alleles from both parents)
- C) Identical in all humans
- D) Found only in RNA

5. CRISPR-Cas9 is a recently developed tool that can:

- A) Only stain chromosomes for microscopy

- B) Target and cut DNA at a specific site, enabling edits to genes
- C) Replace photosynthesis in plants
- D) Prevent all epigenetic change

6. In making a recombinant plasmid, which enzyme “glues” DNA fragments together?

- A) Restriction enzyme only
- B) DNA ligase
- C) RNA polymerase
- D) Helicase

Part B: Fill in the Blank (11 points)

Questions 1–11. Complete each statement with the best biological term from the word bank. Each question is worth 1 point.

Word bank: Anticodon, Codominance, Diploid, Epigenetics, Heterozygous, Methylation, Nondisjunction, Plasmid, Restriction, Transcription, Translation

1. _____ is the process in which RNA polymerase synthesizes RNA using a DNA template.
2. During protein synthesis, ribosomes read mRNA codons and carry out _____ to link amino acids into a polypeptide.
3. On a tRNA molecule, the three-base sequence that base-pairs with a complementary codon on mRNA is called the _____.
4. Adding methyl groups to DNA can help silence a gene without changing the order of A, T, G, and C; this chemical tagging is called DNA _____.
5. The study of changes in gene activity that do not change the DNA sequence is called _____.
6. An organism with two different alleles for a gene (for example, Aa) is described as _____.

7. If chromosomes fail to separate properly during meiosis, the error is called _____ and may produce aneuploid gametes.
8. In the ABO system, type AB blood illustrates _____ because both A and B alleles are fully expressed.
9. Bacterial genes are often cloned using a small circular piece of DNA called a _____.
10. A _____ enzyme cuts DNA at a specific short sequence, creating “sticky” or blunt ends for cloning.
11. Human somatic cells are _____ (2n), containing two sets of chromosomes.
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Part C: Free Response (9 points)

Choose THREE of the following five questions. Answer clearly in the space provided. Each response is worth 3 points.

1. **Information Flow and Regulation.** Trace the path from DNA to a functional protein (name the major steps). Then add one sentence explaining how epigenetics or gene regulation can change whether that gene’s product is actually made in a given cell type—without changing the DNA sequence.
2. **Mitosis vs. Meiosis.** Compare mitosis and meiosis in terms of: number of divisions, number of daughter cells, ploidy (haploid vs. diploid) of the products, and whether the daughter cells are genetically identical to each other. Also state one main purpose of each process.
3. **Inheritance in Words.** Explain the difference between genotype and phenotype. Then describe what incomplete dominance looks like in offspring compared with simple dominant/recessive inheritance (you may use a flower-color example).
4. **Epigenetics vs. Mutation.** Define epigenetics in your own words and contrast it with a DNA mutation. Give one example of an environmental factor that can influence gene

activity through epigenetic-style regulation (diet, stress, exercise, or toxins—pick one and explain briefly).

5. **Biotechnology Toolkit.** Briefly explain what PCR accomplishes and why gel electrophoresis is useful afterward. OR explain how inserting a foreign gene into a bacterial plasmid relates to the universal genetic code (either option is acceptable—state which you chose).

Response 1

Question Number: _____

Response 2

Question Number: _____

Response 3

Question Number: _____

Feedback (Optional)

A. Is there anything else you would like to share regarding your exam preparation, or concepts you found particularly interesting/challenging?

B. What score (or percentage) do you estimate you earned on this exam? How confident are you in that estimate (1–10)?

End of Exam 02